EvidenceBot: A Privacy-Preserving, Customizable RAG-Based Tool for Enhancing Large Language Model Interactions

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Abstract

Large Language Models (LLMs) have become pivotal in reshaping the world by enabling advanced natural language processing tasks such as document analysis, content generation, and conversational assistance. Their ability to process and generate human-like text has unlocked unprecedented opportunities across different domains such as healthcare, education, finance, and more. However, commercial LLM platforms face several limitations, including data privacy concerns, context size restrictions, lack of parameter configurability, and limited evaluation capabilities. These shortcomings hinder their effectiveness, particularly in scenarios involving sensitive information, large-scale document analysis, or the need for customized output. This underscores the need for a tool that combines the power of LLMs with enhanced privacy, flexibility, and usability.

To address these challenges, we present **EvidenceBot**, a local, Retrieval-Augmented Generation (RAG)-based solution designed to overcome the limitations of commercial LLM platforms. EvidenceBot enables secure and efficient processing of large document sets through its privacy-preserving RAG pipeline, which extracts and appends only the most relevant text chunks as context for queries. The tool allows users to experiment with hyperparameter configurations, optimizing model responses for specific tasks, and includes an evaluation module to assess LLM performance against ground truths using semantic and similarity-based metrics. By offering enhanced privacy, customization, and evaluation capabilities, EvidenceBot bridges critical gaps in the LLM ecosystem, providing a versatile resource for individuals and organizations seeking to leverage LLMs effectively.

CCS Concepts

• **Information systems** → **Information retrieval query processing**; • **Security and privacy** → *Systems security*.

Keywords

Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Privacy-preserving AI, Hyperparameter Optimization, LLM Evaluation Metrics

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1 Introduction

LLMs are increasingly used for various tasks, ranging from conversational support in everyday interactions to the extraction of knowledge from extensive documents [9, 15, 36]. Their applications span not only personal use, but also widespread adoption across industries such as e-commerce and retail, education, finance, and healthcare. According to a recent analysis, by 2025, approximately 750 million applications worldwide will integrate LLM, automating almost 50% of digital workflows [32]. Furthermore, the market capitalization of the LLM industry is projected to reach 2,598 million USD by 2030, highlighting the transformative potential of these technologies.

Prominent organizations, including OpenAI [28], Google [14], and Anthropic [2], have developed foundational models such as `GPT-4`, `Gemma`, and `Claude`, respectively. These models are also supported by web-based platforms like ChatGPT (OpenAI) [27], Gemini (Google) [13], and Claude (Anthropic) [3], which allow general users to have LLM capabilities in a wide range of tasks. Despite these platforms' advantages, they are accompanied by significant challenges related to **Cost and Data Privacy**. Most commercial LLM platforms are not freely accessible, which requires costly subscriptions for extended use. Furthermore, user data confidentiality remains a critical concern, as these platforms process queries on remote servers. This data exchange inevitably raises ethical and privacy issues, particularly when handling sensitive information. As reported in a recent study [17], while 67% of organizations use generative AI products that depend on LLMs, only 23% are planning to implement commercial LLMs due to ongoing privacy and ethical concerns.

In addition to cost and privacy challenges, commercial LLM platforms such as `ChatGPT`, `Gemini`, and `Claude` have **Data Upload limitation** as they lack built-in retrieval-augmented generation (RAG) capabilities [12]. When querying multiple large documents, these platforms ingest the entire content, limiting users to a specific document size and restricting the number of documents that can be uploaded. Inclusion of irrelevant data may further confuse the model, leading to suboptimal responses. RAG frameworks offer a solution to this problem by breaking documents into smaller, manageable chunks and retrieving only the relevant portions as context for queries, thus ensuring both efficiency and relevance without overwhelming the model.

In response to the limitations of commercial LLM platforms, several open-source RAG-based tools have emerged that offer alternatives to query large documents. Projects like Kotaemon [8], Realtime-Indexer-QA-Chat [18], and Verba [34] allow users to query LLMs using large documents more effectively. However, these tools also present certain limitations, which can be categorized as follows.

Lack of Parameter Customization: Both open source and commercial LLM platforms typically do not allow users to modify model parameters that play a critical role in tailoring output to specific needs. This lack of customization restricts users' control over the quality and specificity of the generated responses. Furthermore, the operational mechanics of RAG techniques remain opaque in many tools, as users are often unable to modify key RAG-related parameters, such as selecting the embedding model for vector search, adjusting the chunking size of documents, or determining the number of text chunks to consider as context for the RAG model. Users with adequate computational resources often seek the flexibility to experiment with these parameters to optimize the performance of the RAG framework.

Evaluation Constraints: A significant shortcoming of existing platforms is the lack of tools to evaluate model-generated responses against ground truth data. This deficiency prevents users from evaluating the relative performance of different models, an important feature to identify the most appropriate solution for specific tasks.

Given these limitations, the development of a flexible LLM-based platform becomes of paramount importance. In this paper, we introduce EvidenceBot, a local RAG-based solution designed to address the shortcomings of current LLM platforms. EvidenceBot allows users to: 1) experiment with different model- and RAG framework-based parameters for fine-tuning responses; 2) Evaluate and compare the performance of various LLMs against ground truths; 3) Ensure data privacy and overcome constraints related to context size and document processing.

EvidenceBot represents an attempt to systematically address the limitations of commercial LLM platforms, bridging the gap between open-source LLMs and their practical and widespread adoption.

2 Tool Development

2.1 Development of the RAG Pipeline

The RAG pipeline is central to the system. It allows the creation of a knowledge base from input documents and provides context to the model along with the user's prompt [12]. This approach ensures that the model can incorporate relevant information when answering the questions [6].

The architecture of the proposed pipeline is shown in Fig. 1. The pipeline is capable of processing documents in various formats, such as `HTML`, `txt`, `md`, `py`, `pdf`, `csv`, `xlsx`, `docx`. To handle these different file types, the pipeline utilizes the `LangChain` document loader [20]. Once the documents are loaded, they are divided into smaller text chunks (default size = 512 tokens), with the option for the user to adjust the chunk size, thus controlling the number of chunks created for the given set of documents. These text chunks are then converted into embeddings using a preselected embedding model.

Following this, the pipeline leverages `LangChain` [19] to generate semantic indexes for each embedding, facilitating the retrieval and ranking of relevant information based on context and meaning, rather than relying on simple keyword matching [29]. These embeddings and their corresponding semantic indexes are stored in `ChromaDB` [7], a high-performance vector database optimized for efficient similarity search [35].

When a user submits a query, the pipeline converts it into an embedding using the same model as previously used for the documents. Then a semantic search is performed within the vector database, returning the top k most relevant results. The user can specify the value of k within the tool. These relevant text snippets are appended to the user's query and passed to the language model to ensure that the model has the complete context required to generate an accurate response. For compiling and running the language models, we used `Ollama` [26], which is an open-source platform that facilitates the deployment and management of local LLM models.

2.2 Development of the Dashboard

The EvidenceBot dashboard uses `Streamlit`, `HTML`, and `CSS` to create an intuitive, interactive user interface. `Streamlit` works as the primary framework, which enables integration with `Python` logic and language models while managing user inputs, file uploads, and dynamic outputs. `HTML` structures key elements such as the navigation bar, input forms, and chat containers, and `CSS` is used to enhance the visual design, setting background colors, adjusting container layouts, and ensuring responsiveness. The combination of these tools ensures a streamlined user experience.

3 Tool Functionalities

The tool provides two distinct user functionalities, outlined as follows.

3.1 Response Generation

Response generation works in two different modes: `Chat` mode and `Batch` mode. In `Chat` mode, the user can interact with the tool as a chat bot, while in batch mode, the user can pass a set of queries to the system and obtain responses. In both of the modes, the responses are directly stored in a `log.csv` file in a `History` directory of the system. In `Chat` mode, in addition to showing the responses in the front end of the tool, the responses are also stored in the predefined location of the source directory. The system collects timestamps for each query generation and stores these timestamps along with the query responses.

For both modes, the user can set three RAG-based parameters and nine model parameters. The description of the parameters that users can control is provided in Section 3 of the Appendix File [25].

3.2 Response Evaluation

This functionality allows the user to compare the model-generated response with the ground truth/desired response. The system uses four quantitative evaluation metrics inspired by the prior literature [1, 33]. The selected metrics can evaluate the model-generated response while giving the gold annotation as a reference. Among the selected metrics, two are text-similarity-based (*BLEU-4*, and *ROUGE-L*), and the other two are semantic-based (*BERTScore*, and

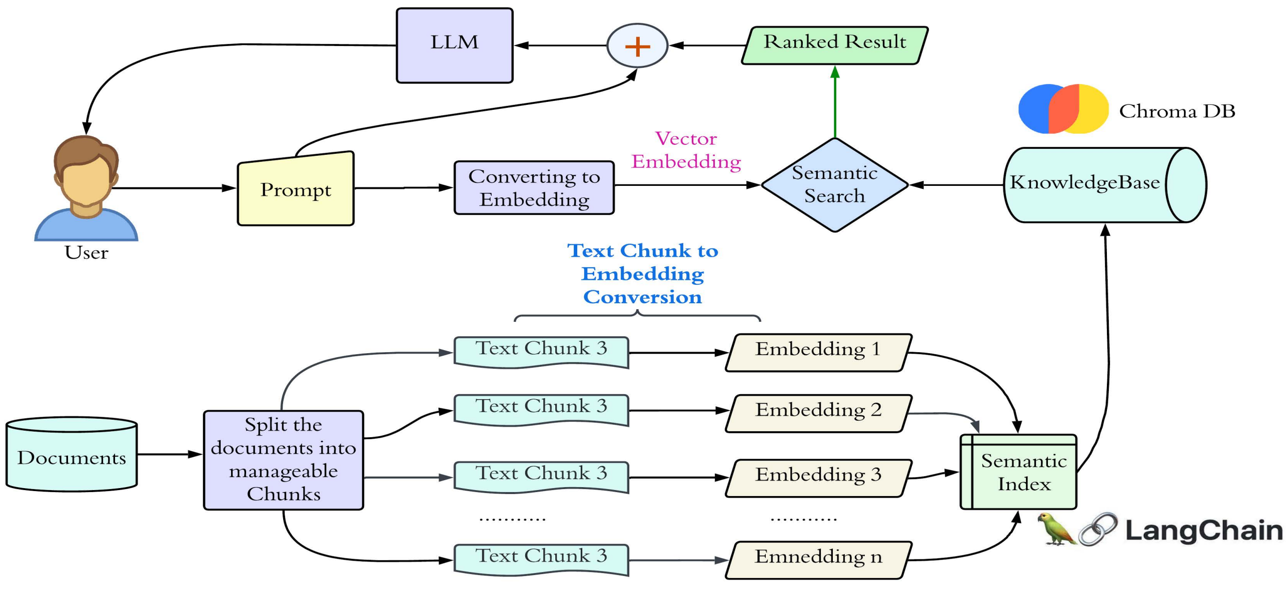


Figure 1: Architecture of the proposed RAG pipeline used in the backend of the tool

METEOR). Semantic-based metrics evaluate text by analyzing meaning and context, considering factors such as coherence and relevance [16]. In contrast, similarity-based metrics focus mainly on lexical overlap or statistical similarity between texts [11].

Short descriptions of the selected evaluation metrics are as follows.

BLEU-4. *BLEU-4* (Bilingual Evaluation Understudie) calculates the geometric mean of the precisions of n-grams up to 4-grams [30]. This metric captures local fluency and adequacy by measuring the overlap of short word sequences (up to 4 words long) between the generated text and reference translations.

ROUGE-L. *ROUGE-L* (Recall-Oriented Understudy for Gisting Evaluation - Longest Common Subsequence) measures the longest common subsequence between the generated output and the reference text [21]. Unlike n-gram-based metrics, ROUGE-L captures sentence-level structure and allows for word-order flexibility.

BertScore. *BERTScore* leverages pre-trained BERT embeddings to assess the semantic similarity between generated text and reference text [37]. In this tool, we utilized the `BERT-base-uncased` [5] variant of BERT, which has 12 layers and 768 hidden units; "uncased" means that the model does not differentiate between upper-case and lowercase letters.

METEOR. *METEOR* (Metric for Evaluation of Translation with Explicit ORdering) incorporates linguistic features to capture semantic equivalence and structural adequacy [4]. Its flexible matching strategy accounts for exact matches, stemming, synonymy, and paraphrasing; which provides a more nuanced evaluation of semantic similarity.

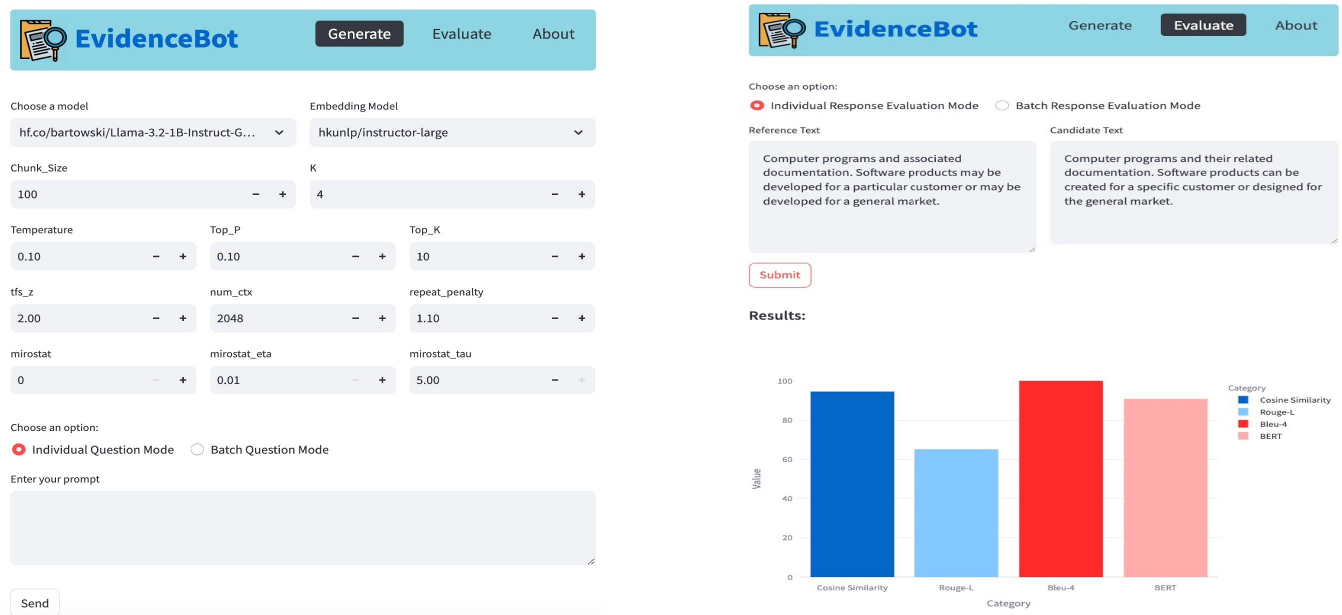
4 Use Cases

The proposed tool offers three primary use cases, as described below:

Use Case I: Privacy-Preserving Customizable RAG Pipeline. The proposed tool provides a local solution to address privacy and ethical concerns associated with the use of LLMs to process confidential data. At its core lies a RAG pipeline, which enables users to query LLMs using a large set of documents. When a user inputs documents and a text query, the pipeline identifies relevant text chunks from the provided documents and appends only the relevant chunks (as specified by the user in the UI) as context for the model. This ensures that the extracted context is directly aligned with the user query, preventing unnecessary data from overloading the model.

The tool also offers flexibility for users to configure the size of text chunks and the number of chunks to pass as the context to the model. For instance, when using models with larger context windows, users can opt for larger chunk sizes and increase the number of extracted chunks. In contrast, for models with smaller context windows, users can select smaller fragments and limit the number of pieces extracted. This privacy-preserving customizable RAG pipeline allows users to experiment with different parameters while ensuring that confidential data does not leave their local machine.

Use Case II: Experimenting with Model Parameters The performance of LLMs can vary significantly depending on the model parameters [22], which dictate how the model generates responses. For example, adjusting the temperature parameter can influence the model's creativity; increasing the temperature produces more diverse outputs, while decreasing it makes the responses more deterministic [31]. Other parameters similarly affect the response generation process.



(a) (b)

Figure 2: Screenshots of EvidenceBot: (a) Response Generation, (b) Response Evaluation

The tool allows users to experiment with various model parameters to observe their impact on responses. Since determining the optimal parameters for a specific use case is not a straightforward process, this feature enables users to test different configurations and identify the best set of parameters for their specific needs.

Use Case III: Evaluating Responses from Various LLMs. The tool includes functionality to evaluate LLM-generated responses against ground truths using metrics that assess both semantics (e.g. *METEOR*, *BERTScore*) and text similarity (e.g., *BLEU-4*, *BERTScore*). When users have ground-truth data, they can compare LLM responses to ground truth by analyzing these evaluation metrics.

The tool also provides a graphical representation of the evaluation metrics. Additionally, the evaluation module supports batch processing, allowing users to upload CSV files containing multiple LLM responses and corresponding ground-truth data. This enables the evaluation of a large number of samples.

5 Limitations & Conclusion

The developed tool, while robust in many aspects, has some limitations. Currently, it can process data only in textual formats, which means it cannot extract information from audio, image, or video inputs, nor can it generate responses in non-textual formats. Extending the tool to handle both input and output of non-textual data would be a significant enhancement for future development. Furthermore, the tool relies on the `Ollama` server on the back end for LLM interactions, which supports only models in GGUF format. As a result, open source models in other formats, such as those available on *HuggingFace* [10], cannot be directly used with the tool. Addressing this limitation to enable compatibility with diverse model formats would further expand the flexibility and adoption of the tool.

Despite these constraints, `EvidenceBot` offers several unique features that make it a valuable resource. Its customizable RAG pipeline ensures efficient processing of large document sets by extracting and appending only the most relevant text chunks as context for the model. This approach minimizes the risk of exceeding the model's context window while maximizing the response relevance. The tool also supports batch query execution, enabling users to process multiple queries simultaneously. Additionally, the ability to evaluate LLM-generated responses against ground truths using various metrics allows users to assess the effectiveness of different models. Furthermore, the tool's capability to experiment with hyperparameter configurations empowers users to fine-tune LLM performance for their specific needs.

Considering the absence of open-source LLM bots with such integrated functionalities, EvidenceBot addresses a critical gap and provides a versatile solution for a wide range of applications. With its ability to handle large-scale document analysis, evaluation capabilities, and parameter experimentation, the tool holds immense potential for both general and specialized use cases in the future.

Tool & Data Availability

The data and code utilized in this study are publicly available at [24]. A screencast demonstrating the tool can be accessed at [23]. Additionally, the appendix file is available at [25].

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